PROJECT DESIGN PHASE-I

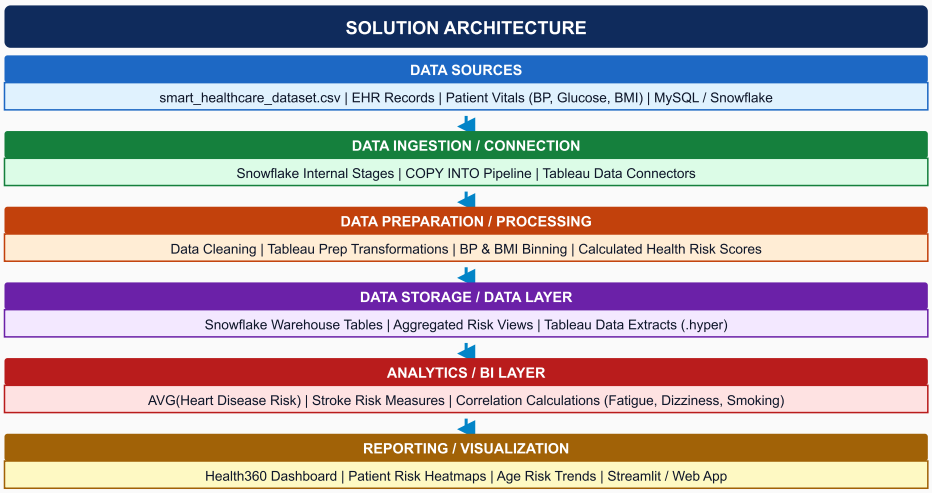
Solution Architecture

Date	28 October 2026	Maximum Marks	4 Marks
Team ID	EN25CS3011166	Project Name	Health360: Cardiovascular Risk Analytics

Solution Architecture: Solution architecture bridges the gap between healthcare data and analytical decision-making. Its goals are to:

- Find the best tech solution to solve cardiovascular risk visualization and early diagnostic detection.
- Describe the multi-tier structure, ETL pipelines, and visual analytics layers to stakeholders.
- Provide technical specifications for dynamic filtering across BP, BMI, Glucose, Fatigue, and Smoking parameters.

Solution Architecture Diagram:



Component Description Table:

Component Name	Description / Role in Architecture	Technology Used
Data Sources	Stores raw physiological dataset (`smart_healthcare_dataset`) with metrics like Age, BMI, Blood Pressure, Cholesterol, Fatigue, and Smoking.	**CSV, MySQL, Snowflake Data Warehouse**
Data Ingestion / Connection	Loads bulk clinical data into staging environments and establishes live/extract data connections for real-time visualization.	**Snowflake Stages, COPY INTO, Tableau Connectors**
Data Preparation / Processing	Cleans missing values, calculates binned ranges (BP 113.40-119.07+), and formulates calculated fields like AVG(`Heart Disease`).	**Tableau Prep, Python (Pandas), SQL Views**
Data Storage / Data Layer	Maintains high-performance columnar data extracts (.hyper) and aggregated database views for instant dashboard query execution.	**Snowflake Analytics Warehouse, Tableau Extract**
Analytics / BI Layer	Executes business logic for cardiac/stroke risk scoring, cross-variable correlation (Fatigue v/s Stroke Risk, Chest Pain v/s Dizziness).	**Tableau Desktop, Snowflake SQL Engine**
Reporting / Visualization	Delivers 8 interactive diagnostic panels: Patient Risk Heatmap, Age Risk Trends, Alcohol & Smoking Risk, and BMI vs Heart Risk.	**Tableau Public Web, Streamlit, HTML5/CSS3**